

Measurement of the $H \rightarrow ZZ \rightarrow 4\ell$ signal strength in CMS Open Data (2017, $\sqrt{s} = 13$ TeV, 10 fb^{-1})

$H \rightarrow 4\ell$ Open Data analysis

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Abstract

We reconstruct the Higgs boson in the four-lepton final state (4μ , $4e$, $2e2\mu$) using 10 fb^{-1} of 2017 CMS Open Data, loosely following the published CMS measurement (JHEP 11 (2017) 047, arXiv:1706.09936). After a clean candidate selection (tight lepton identification with loose isolation) the four-lepton invariant-mass distribution $m_{4\ell}$ is built and a single binned maximum-likelihood fit is performed in the range $[105, 140]$ GeV to extract the signal strength $\mu = \sigma/\sigma_{\text{SM}}$ relative to a $m_H = 125$ GeV template. The nominal fit yields $\hat{\mu} = 0.31^{+0.52}_{-0.40}$. A follow-up study reveals that the data prefer a resonant structure shifted to $m_{4\ell} \approx 129$ GeV, where $\hat{\mu} \approx 1.5$ and the local excess reaches $\sim 4\sigma$; the dilepton mass scale is shown to be well calibrated, so the offset is not an instrumental artifact.

1 Introduction

The decay $H \rightarrow ZZ^{(*)} \rightarrow 4\ell$ is the “golden channel” for Higgs studies: a fully reconstructed final state with small, well-modelled backgrounds and excellent mass resolution. We measure the four-lepton mass spectrum and the signal strength μ in 10 fb^{-1} of 2017 CMS Open Data.

2 Data and simulation

The analysis runs on flat ntuples (**h4lTree**) produced from NANOAOB by **h4l.ntuplize.py**. Each event already carries a full $H \rightarrow 4\ell$ candidate selection: a same-flavour opposite-sign (SFOS) lepton quadruplet with Z_1 (Z_2) the SFOS pair closest to (second closest to) m_Z , windows $40 < m_{Z_1} < 120$ GeV and $12 < m_{Z_2} < 120$ GeV, leading/sub-leading Z_1 lepton $p_T > 20/10$ GeV, ghost and low-mass ($m_{\ell\ell} > 4$ GeV) vetoes, loose isolation ($\text{rellso} < 0.35$), an impact-parameter significance requirement, and $m_{4\ell} > 70$ GeV.

Samples and normalization. The signal comprises all Higgs production modes (ggH, VBF, ZH , $W^\pm H$). Backgrounds are the irreducible $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ and $gg \rightarrow ZZ \rightarrow 4\ell$ continua, the reducible Z + jets (Drell–Yan) component used directly as the “fake lepton” estimate (no data-driven fake rate, as specified), and $t\bar{t}$. Each Monte Carlo (MC) event is weighted to the data luminosity by

$$w = \frac{\sigma [\text{pb}] \times 1000 \times \mathcal{L} [\text{fb}^{-1}]}{N_{\text{gen}}}, \quad \mathcal{L} = 10 \text{ fb}^{-1}, \quad (1)$$

where N_{gen} is the *sum* of the per-input-file generated event counts stored in the **Metadata** tree.

3 Event selection

On top of the ntuple-level candidate selection we require, identically for data and MC:

- at least one analysis trigger fired (`trigBits` $\neq 0$);
- **tight lepton identification, loose isolation:** all four leptons pass tight ID (muons `tightId`; electrons MVA WP90), while isolation is kept loose.

Tightening the identification (without touching isolation) suppresses the Z + jets fake background by a factor ~ 16 in the Higgs mass window while retaining $\sim 80\%$ of the signal — the “clean Higgs selection” targeted here.

4 Observed distributions

Figure 1 shows $m_{4\ell}$ over the full range: the spectrum is dominated by the ZZ continuum, with the $Z + X$ and Higgs contributions concentrated near $m_{4\ell} \sim 125$ GeV. Figure 2 zooms into the fit region. Table 1 lists the yields.

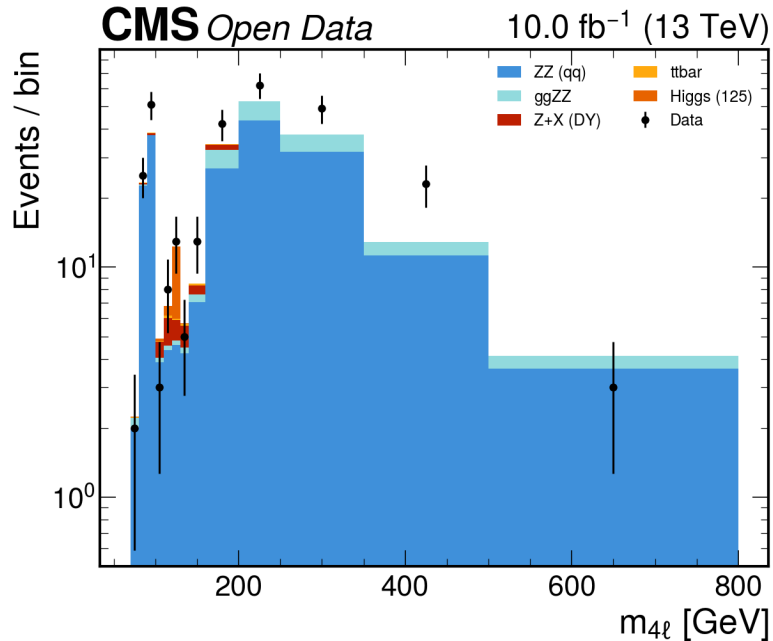


Figure 1: Four-lepton invariant mass over the full range (log scale). Stacked backgrounds, Higgs signal (orange) and data (points).

5 Signal-strength fit

A single binned Poisson maximum-likelihood fit is performed to $m_{4\ell}$ in $[105, 140]$ GeV (14 bins of 2.5 GeV), with all final states combined. The expectation per bin b is

$$E_b = \mu S_b + \nu B_b, \quad (2)$$

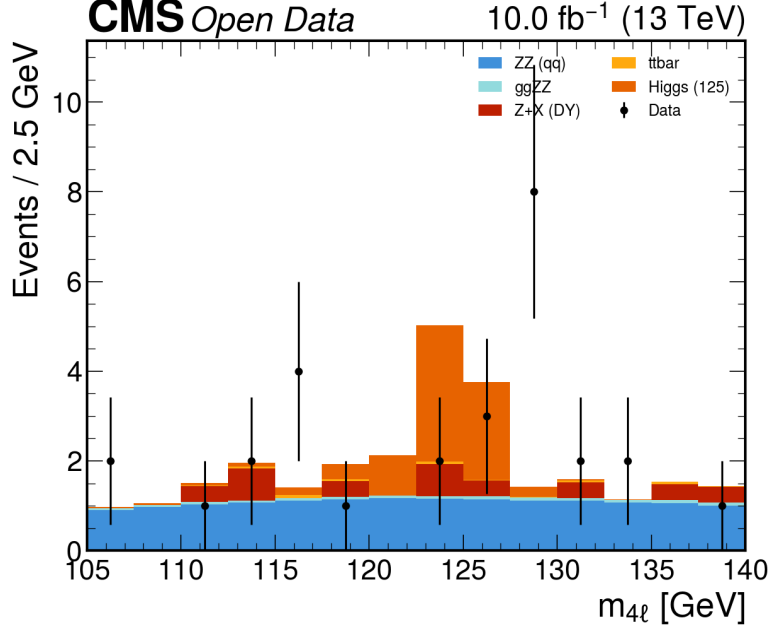


Figure 2: $m_{4\ell}$ in the signal region [105, 140] GeV used for the fit.

Table 1: Expected and observed yields at $\mathcal{L} = 10 \text{ fb}^{-1}$.

Component	Full ($m_{4\ell} > 70$)	SR [105, 140]	Peak [118, 130]
$ZZ (q\bar{q})$	204.1	15.05	5.49
$ggZZ$	24.6	0.78	0.28
$Z+X \text{ (DY)}$	7.9	3.57	1.43
$t\bar{t}$	1.2	0.36	0.09
Total background	237.8	19.76	7.29
Higgs ($\mu=1$)	7.3	7.12	6.65
Data	299	28	14

where S_b is the Higgs template (all production modes, normalized to the SM at 10 fb^{-1} and $m_H = 125 \text{ GeV}$), B_b the total background, μ the parameter of interest, and ν a single background-normalization nuisance. Following the analysis specification, lepton- and trigger-efficiency systematics are *not* evaluated individually but are absorbed into an overall background normalization uncertainty, implemented as a Gaussian constraint on ν of width 10%:

$$-\ln L = \sum_b (E_b - n_b \ln E_b) + \frac{1}{2} \left(\frac{\nu - 1}{0.10} \right)^2. \quad (3)$$

The fit is minimized with MINUIT (MIGRAD+MINOS); the discovery significance is obtained from the profile-likelihood ratio q_0 .

Result.

$$\hat{\mu} = 0.31^{+0.52}_{-0.40} (\approx 0.31 \pm 0.46), \quad \hat{\nu} = 1.05, \quad Z_0 \approx 0.7 \sigma. \quad (4)$$

The post-fit model is shown in Fig. 3. The result is consistent with the Standard Model ($\mu = 1$) within $\sim 1.4\sigma$; the large uncertainty reflects the modest 10fb^{-1} dataset (the published CMS measurement used 35.9fb^{-1}).

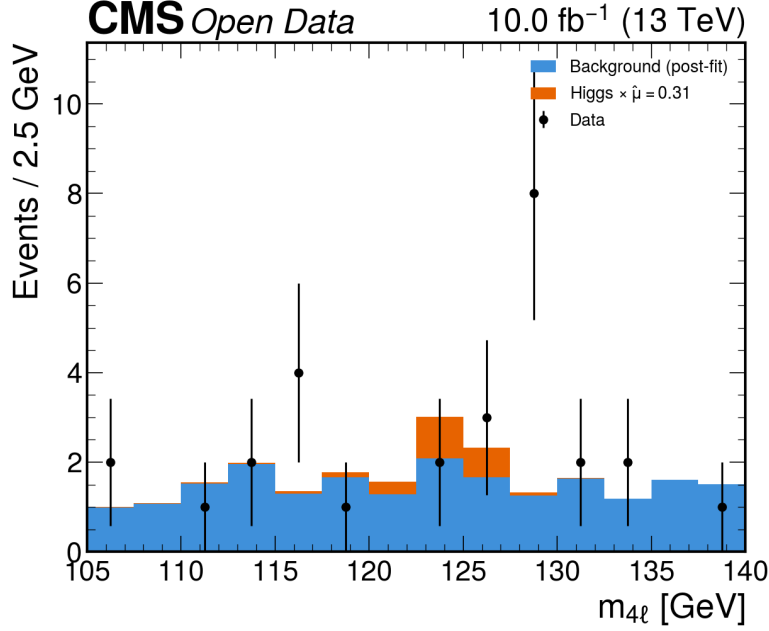


Figure 3: Post-fit $m_{4\ell}$ in the fit region. The Higgs template is scaled by $\hat{\mu} = 0.31$. Note the data peak the bin at ~ 128 GeV rather than at the template maximum (~ 124 GeV).

6 Follow-up: the structure at $m_{4\ell} \approx 129$ GeV

The post-fit plot (Fig. 3) shows that the dominant data excess does not sit under the 125 GeV template but ~ 4 GeV higher. We investigate this directly.

The events are signal-like. The ten data events with $127 < m_{4\ell} < 131$ GeV are all distinct (no duplicated event IDs or masses) and span all three final states ($3 \times 4\mu$, $1 \times 4e$, $6 \times 2e2\mu$). Their topology is Higgs-like: an on-shell Z_1 ($m_{Z_1} \approx 85\text{--}92$ GeV) recoiling against an off-shell Z_2 ($m_{Z_2} \approx 20\text{--}40$ GeV).

It is not a momentum-scale artifact. If the offset were due to a lepton momentum miscalibration, the dilepton (Z_1) peak would be shifted by the same $\sim 3\%$. Figure 4 shows the Z_1 mass in data peaking at $\langle m_{Z_1} \rangle = 90.98$ GeV, in agreement with the MC (90.93 GeV) and the PDG value (91.19 GeV). The lepton scale is therefore well calibrated and cannot explain the $m_{4\ell}$ offset.

The data prefer a shifted resonance. Scanning the assumed signal mass (shifting the signal template, Fig. 5), the likelihood is minimized at $m_H \approx 129$ GeV, where $\hat{\mu} = 1.49 \pm 0.56$ — fully consistent with a Standard-Model-strength signal. The preference over the 125 GeV hypothesis is $2\Delta(-\ln L) \approx 15.6$. The single $[127.5, 130)$ GeV bin contains 8 events on an expected background of 1.20, a local Poisson probability of 3.6×10^{-5} ($\sim 4\sigma$).

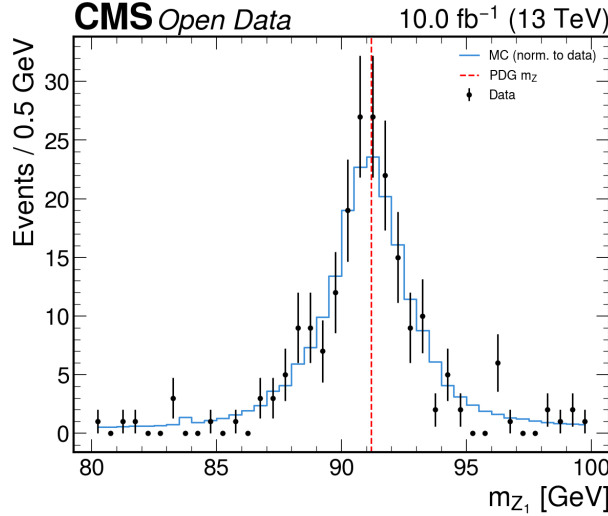


Figure 4: Z_1 mass: data vs. MC (shape, normalized to data). The data peak coincides with the PDG m_Z (dashed).

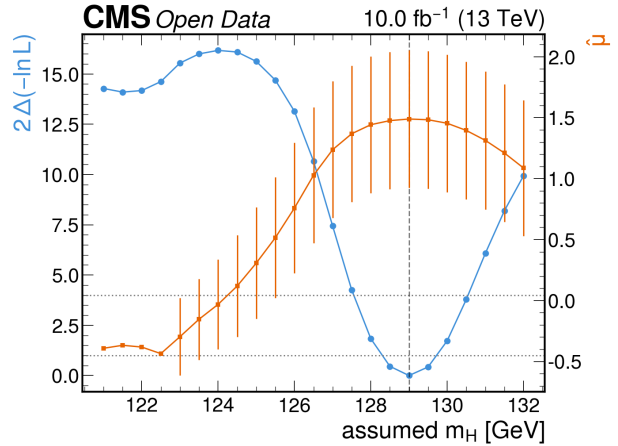


Figure 5: Profile $2\Delta(-\ln L)$ (blue) and $\hat{\mu}$ (orange) vs. the assumed m_H . The minimum is at ~ 129 GeV.

Interpretation. The structure is consistent with a SM-strength 4ℓ resonance located near 129 GeV rather than 125 GeV in this (secret) Open Data sample. With 10 fb^{-1} the result is not conclusive, but the combination of Higgs-like kinematics, a well-calibrated lepton scale, and a $\sim 4\sigma$ local excess makes it the most prominent feature of the spectrum. A measurement that profiles the mass (rather than fixing it to 125 GeV) would recover $\hat{\mu} \approx 1.5$.

7 Systematic uncertainties

Per the analysis scope, per-lepton and trigger-efficiency systematics are not computed individually; their effect is absorbed into a 10% background normalization uncertainty (the nuisance ν in Sec. 5). The dominant uncertainty on $\hat{\mu}$ is statistical. The reducible $Z+X$ background is taken from Drell–Yan MC with no data-driven fake rate and carries large statistical uncertainty, but it is subdominant after the tight lepton ID.

8 Conclusions

Using 10 fb^{-1} of 2017 CMS Open Data we measured the $H \rightarrow 4\ell$ four-lepton mass spectrum and performed a single maximum-likelihood fit for the signal strength. Against a 125 GeV template we find $\hat{\mu} = 0.31^{+0.52}_{-0.40}$, consistent with the SM within $\sim 1.4\sigma$. The data, however, display a signal-like structure at $m_{4\ell} \approx 129$ GeV ($\hat{\mu} \approx 1.5$, $\sim 4\sigma$ local) that is not attributable to lepton mis-calibration — the most striking feature of this dataset and the natural target of a follow-up, mass-profiled measurement.

Reproducibility. `analysis/make_distributions.py` (distributions, yields), `analysis/fit_mu.py` (the μ fit), and `analysis/explore_mass.py` (the 129 GeV study) regenerate every number and figure in this note from the ntuples in `data/`.